Gaussian gauge blocking: exact covariance, decay and the nonlinear threshold

For the specified linearized four-dimensional $SU(3)$ block average, hard conditioning gives an exact Gaussian covariance and the explicit tube estimate $-\Delta|_{\ker W} \geq 4/9$. A distinct soft constraint has gap $2/9$ for coefficient $\alpha \geq 130$ and the explicit decay criterion below. These are Gaussian statements. No normalized nonlinear polymer expansion or physical continuum gap is proved by them.

The coarse variable already averages **two-link paths**. Its angle is $\theta_c = W\theta$, without a second factor of two. Likewise, an ℓ^∞ kernel bound requires the corresponding lattice shell sum, rather than the smaller product sum for ℓ^1 distance. These corrections change the former normalization and numerical threshold estimates. The surviving nonlinear estimate is a conditional scaling diagnosis, not a certified Kotecký–Preiss region or a theorem excluding other methods.

Conventions and gauge-compatible averaging

On $\mathbb{Z}^4$ use lattice units $a = 1$, Hermitian generators with $\operatorname{tr} T^a T^b = \delta^{ab}/2$, link angles $\theta = aA$ and $U_\ell = e^{i\theta_\ell}$. The dimensionless coupling obeys $g^2 = \hbar g_{\mathrm{cl}}^2$ when a classical normalization is supplied. Then

$$\frac{S_W}{\hbar} = \frac{2}{g^2}\sum_p (N - \operatorname{Re}\operatorname{tr} U_p) = \frac{1}{2g^2}\|d\theta\|^2 + O(\theta^3).$$

The cubic/quartic terms, Haar correction and nonlinear averaging belong to the interacting remainder. The action-cost treatment distinguishes an action lower bound from normalized interacting rarity.

For blocks B of side two define

$$(W\theta)_{B,\mu} = \frac{1}{16}\sum_{x\in B}(\theta_{x,\mu} + \theta_{x+\hat{\mu},\mu}), \qquad (W_0\lambda)_B = \frac{1}{16}\sum_{x\in B}\lambda_x.$$

The path telescopes, giving the exact intertwining

$$Wd = \bar{d}W_0, \qquad (Wd\lambda)_{B,\mu} = (W_0\lambda)_{B+2\hat{\mu}} - (W_0\lambda)_B.$$

Thus a fine linearized gauge transformation induces a coarse one. This is a linearized identity, not a construction of the nonlinear group-valued map.

The tube for (B,μ) has three μ-layers of eight sites, weighted $(1,2,1)$. Its total weight is 32; the normalization $1/16$ makes a constant fine angle map to twice that angle, as a coarse link of length $2a$ requires. Every fine link lies in one or two tubes of its direction.

Exact conditional covariance and the tube gap

Adding $\|d^*\theta\|^2/(2g^2)$ gives 32 scalar Feynman-gauge problems. Use the infinite-lattice Green function in four dimensions, or fix the constant modes in a finite-volume version before writing inverses. Put $G = (-\Delta)^{-1}$ and $K = \ker W$. The minimizer with $Wf = \bar{f}$ is

$$H\bar{f} = GW^T(WGW^T)^{-1}\bar{f}.$$

Variational orthogonality gives

$$\langle f, -\Delta f\rangle = \langle H\bar{f}, -\Delta H\bar{f}\rangle + \langle \xi, -\Delta\xi\rangle, \quad \xi = f - H\bar{f} \in K.$$

Consequently

$$C_{\mathrm{fl}} = g^2[G - GW^T(WGW^T)^{-1}WG] = g^2(P(-\Delta)P)^{-1}|_K,$$

$$\bar{S}_2(\bar{f}) = \frac{1}{2g^2}\langle \bar{f}, (WGW^T)^{-1}\bar{f}\rangle,$$

where P projects onto K. Linearized gauge-invariant observables have the same law in Feynman and Landau gauges: their difference is a pure gauge, and $Wd = \bar{d}W_0$ makes its coarse image pure gauge as well.

Tube Poincaré bound. For every $f \in K$,

$$\langle f, -\Delta f\rangle \geq \frac{4}{9}\|f\|^2.$$

To prove it, one tube is a $2\times 2\times 2\times 3$ graph with first nonzero Laplacian eigenvalue 1. For its plain mean m,

$$\|f-m\|_T^2 \leq \sum_{\text{bonds}\subset T} |\partial f|^2.$$

The weighted mean vanishes; with $\bar{w} = 4/3$ and $\sum_T (w-\bar{w})^2 = 16/3$,

$$m = -\frac{1}{32}\sum_T (w-\bar{w})(f-m), \qquad 24m^2 \leq \frac{1}{8}\|f-m\|_T^2.$$

Hence $\|f\|_T^2 \leq (9/8)\sum_{\text{bonds}\subset T}|\partial f|^2$. Summing tubes covers every site and at most doubles each bond, proving $\|f\|^2 \leq (9/4)\|\partial f\|^2$. Alternating layers or transverse alternating modes have energy $4\|f\|^2$ and satisfy the constraints, so the optimal constant lies in $[4/9, 4]$.

In particular,

$$\|C_{\text{fl}}\| \leq \frac{9}{4}g^2, \qquad (\mathbb{E}\xi_\ell^2)^{1/2} \leq \frac{3}{2}g.$$

For one centered Gaussian colour component this implies

$$\Pr(|\xi_\ell| \geq \eta) \leq 2e^{-2\eta^2/(9g^2)}.$$

It bounds that Gaussian fluctuation only; a nonzero background shifts the threshold, and nonlinear reweighting requires its own normalized estimate.

For slowly varying fields, $Wf \simeq 2f$ and coarse differences are $\bar{\partial}(Wf) \simeq 4\partial f$. The coarse site density is $1/16$. These factors give the same Maxwell coefficient $1/(2g^2)$ in four dimensions, with relative momentum corrections of order $\bar{k}^2a^2$. The variable is $\theta_c = \bar{f}$, not $2\bar{f}$. This is the leading Gaussian coefficient, with no nonlinear running claim.

Hard conditioning and soft-constraint decay

For hard conditioning, P is nonlocal. Along the elongated direction,

$$WW^T = \frac{48 + 8(S+S^{-1})}{256}, \qquad \operatorname{spec}(WW^T) \subset [1/8, 1/4].$$

Conjugating $P(-\Delta)P + (4/9)(1-P)$ by exponential weights motivates the earlier hard-constraint rate near 7×10^{-4}. The displayed projection calculation used $\|P_\kappa - P\| \leq 10\kappa + O(\kappa^2)$ without a finite-κ remainder bound. That number is an estimate, not a certified explicit rate in this treatment. In particular a rate for the following different operator cannot silently certify C_{fl}.

Let $Q = W$ and $A_\alpha = -\Delta + \alpha Q^TQ$. Write $f = f_K + f_\perp$. The tube bound, $\|-\Delta\| \leq 16$ and $\|Qf_\perp\|^2 \geq \|f_\perp\|^2/8$ give

$$\langle f, A_\alpha f\rangle \geq \frac{2}{9}\|f_K\|^2 + (\alpha/8 - 16)\|f_\perp\|^2 \geq \frac{2}{9}\|f\|^2 \quad (\alpha \geq 130).$$

The elementary first inequality uses $\|\partial(f_K+f_\perp)\|^2 \geq \frac{1}{2}\|\partial f_K\|^2 - \|\partial f_\perp\|^2$. The soft operator has range two in the elongated coordinate and one in each other coordinate. Since Q^TQ has

nonnegative entries and row/column sum 1/4, conjugation in any one coordinate changes it in norm by at most $(e^{2\kappa}-1)/4$. Therefore

$$2\sinh\kappa+\frac{\alpha}{4}(e^{2\kappa}-1)\le\frac{1}{9}\quad\Longrightarrow\quad|A_\alpha^{-1}(x,y)|\le 9e^{-\kappa|x-y|_\infty}.$$

Choose the coordinate attaining $|x-y|_\infty$ for the matrix-element bound. At $\alpha=130$, $\kappa=10^{-3}$ is a conservative explicit choice; for larger α use the displayed criterion with that value. The former 1.7×10^{-3} and its uniform use for all $\alpha\ge130$ were not justified by that inequality.

Dimock, *Rev. Math. Phys.* **25** (2013) 1330010, §2.4 (arXiv passage read in the source work), treats soft Green functions by local Neumann inverses and a random-walk parametrix with sufficiently large cubes. That passage gives parameter-dependent constants, not the missing numerical hard rate. Balaban, *CMP* **95** (1984) 17, is retained at metadata level.

Folded symbol and the unproved analytic strip

For hard conditioning, coarse momentum $\bar k\in[-\pi/2,\pi/2]^4$ couples $k=\bar k+\pi n$, $n\in\{0,1\}^4$. Up to a common normalization,

$$w(k)=\prod_{\nu\neq\mu}\cos^2(k_\nu/2)\,\cos^4(k_\mu/2),\qquad\hat k^2=4\sum_\nu\sin^2(k_\nu/2),$$

$$R(\bar k)=\sum_{n\neq0}\frac{w(\bar k+\pi n)}{\hat k^2(\bar k+\pi n)},\qquad C_{00}(\bar k)\;\propto\;\frac{R(\bar k)}{w(\bar k)+\hat k^2(\bar k)R(\bar k)}.$$

The original massless pole cancels. Here w denotes the analytic extension of the real-momentum squared weight, not an absolute square at complex momentum. In an axial strip, the flipped denominators have positive real part for $\cosh\kappa<2$, and the unfolded weight has modulus at least 1/32. The remaining denominator must still be shown zero-free. An order-one strip is a proposed improvement, not an established decay rate; all folded matrix entries, rather than only this diagonal, need control.

What the remainder estimate actually says

A cubic Wilson vertex has the schematic size C_3g, with C_3 formerly estimated as order 10–30 from $d\xi$ of size $6g$, ξ^2 of size $9g^2/4$ and the colour contraction. Quartic and Haar corrections start at order g^2. These size estimates do not prove a connected expansion. If a valid tree interpolation bounds each edge by a specified covariance sum B, and a rooted plaquette entropy bound $(20e)^{n-1}$ is applicable, the schematic smallness test is $20e^2C_3gB<1$.

For an **assumed** bound $|C(x,y)|\le Ag^2e^{-\kappa|x-y|_\infty}$, the correct unweighted sum is

$$B:=\sup_x\sum_y|C(x,y)|/g^2\le A\,\mathcal{S}_4(\kappa),$$

$$\mathcal{S}_4(\kappa)=1+\sum_{n\ge1}(64n^3+16n)e^{-\kappa n}=1+\frac{64q(1+4q+q^2)}{(1-q)^4}+\frac{16q}{(1-q)^2},\quad q=e^{-\kappa}.$$

Indeed the ℓ^∞ shell has $(2n+1)^4-(2n-1)^4=64n^3+16n$ sites. Thus $\mathcal{S}_4\sim384\kappa^{-4}$. The earlier $\coth(\kappa/2)^4$ is the sum for ℓ^1 distance and is too small here. A weighted polymer norm also spends part of the available decay rate; B as defined above is unweighted.

Under those extra expansion hypotheses one obtains the illustrative sufficient test

$$g<[20e^2C_3A\mathcal{S}_4(\kappa)]^{-1},$$

whose small-rate scaling is $g=O(\kappa^4)$. The old numerical table mixed norms and different covariances and is superseded by this formula. Even a correct sufficient estimate far from order-one coupling proves no optimality barrier for other organizations. The interacting numerator, normalization, large fields and iteration remain open in this scheme. The later retained-barrier treatment has its own weak-strip domain and does not fill these gaps by substitution.

Consequence for STATE

The maintained output is exact linearized blocking, the 4/9 tube gap and the specified soft-constraint decay criterion. Gaussian scale covariance, nonlinear running, normalized rarity and physical spectral transfer remain different assertions. The nonlinear threshold is a conditional diagnostic; no new research is initiated by correcting its normalization and scope.